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Removal and service of the HP08568-60118 attenuator as found in the HP 8566/8568 spectrum analyzer.

In the HP08568-60118, there are 3 attenuator pads (10 dB, 20 dB, 40 dB) which are switched in and out of line by solenoids to deliver 0 to 70dB attenuation in 10 dB steps.

These attenuator modules are now quite old and sometimes fail to change attenuation as expected. Generally, this can be traced to perished o-rings that are part of the mechanism that selects the attenuation steps.

The solenoids drive plungers that are held in place by o-rings. The plungers push switch leaves in and out to change attenuation. If an o-ring perishes, the plunger may not be able to reliably push the switch leaves into place.

The o-rings must be replaced to return the attenuator to full service.

Completion time

This activity could easily take 5 to 6 hours even if everything goes right. You may encounter issues along the way which will, of course, take extra time to resolve.

Required tools

Along with a standard toolkit, you will need the following:

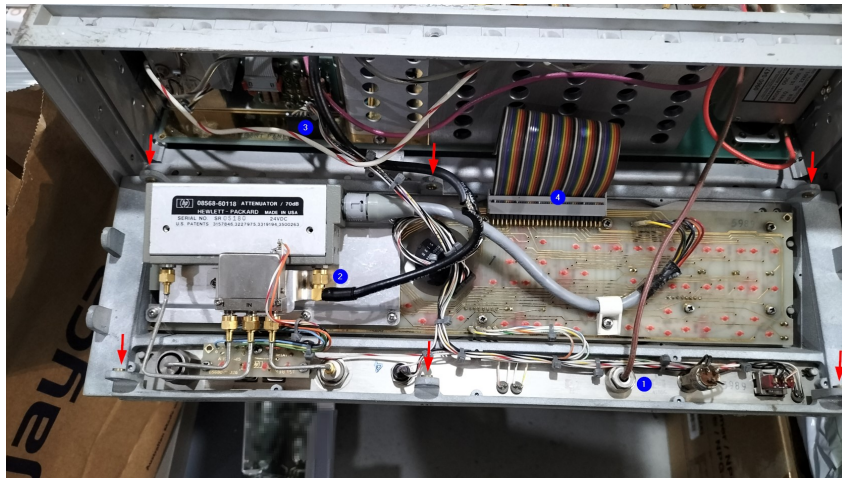
- SMA wrench
- 5/64" Allen key
- Stainless steel (non magnetic) tweezers.
- 24V supply, current limited to ~ 50mA.
- 12 x 2mm O.D., 1mm I.D, 0.5mm thick o-rings.
- A range of containers to hold screws and other parts you may remove during the activity.
- HP 435 power meter or similar for testing.

Disclaimer

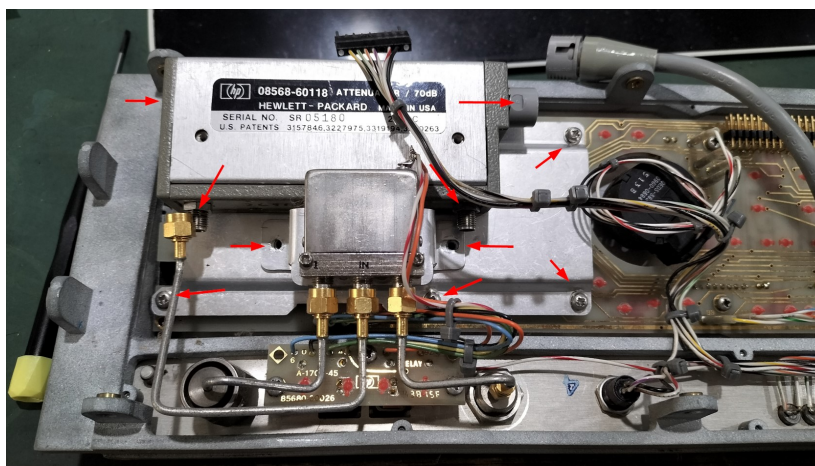
Do not start this task unless you are really sure you are capable of doing the work.

Although low risk, this advice is offered as is with no guarantees and with the understanding that it may contain errors. Should you break your equipment, remember who took the covers off. I am not responsible for breaking your equipment.

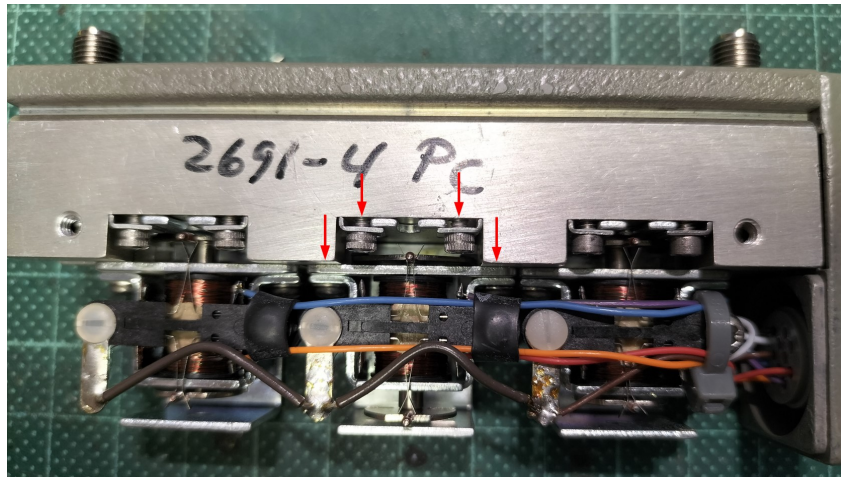
The attenuator is mounted to the rear of the front panel of the 8566/68.



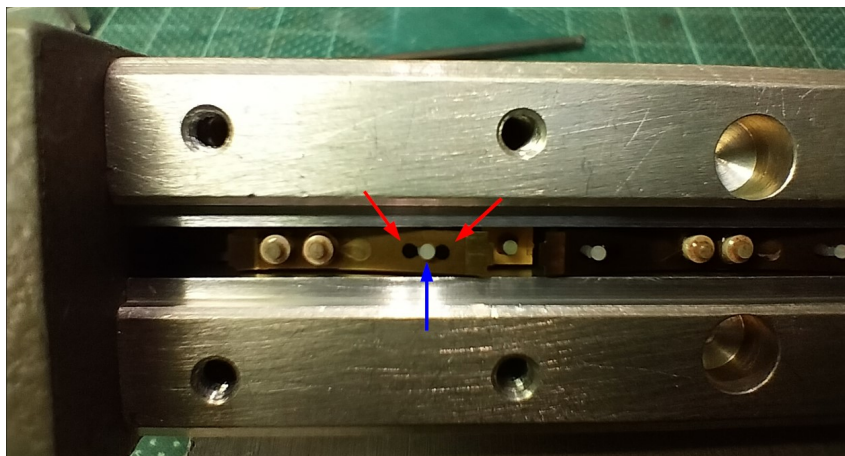
- Remove the 6 screws indicated by the red arrows and swing the panel forward.
- Remove the lock nut from the front of the 20 MHz BNC connector and lay the cable inside the cabinet (1). Disconnect the black coax from the attenuator (2), disconnect the front panel supply cable (3), and the ribbon cable (4). The panel can now be removed from the cabinet.
- The attenuator power comes via a tight fitting “Viking” plug. Squeeze the corrugations on either side of the connector housing and pull back. This requires some effort but with care and patience and it can be removed. **Don't force it; remain in control as you pull the connector. A sudden release of energy could damage the connector or cable.**
- Undo the coax connections from the attenuator, remove the mounting screws holding the coax relay and undo the 5 mounting screws holding the attenuator plate in place.



- Remove the two screws holding the attenuator to the plate and remove the smaller endplate from the attenuator then slide the cover plate length-ways off the assembly. **Do not lever the cover plate off or the solenoids may be damaged.**
- Examine the solenoid assembly carefully. The 3 solenoids are each held in place with 4 screws respectively, indicated by red arrows. **Be careful, because once you remove all the screws, the solenoids are only held together by their connecting wires.**



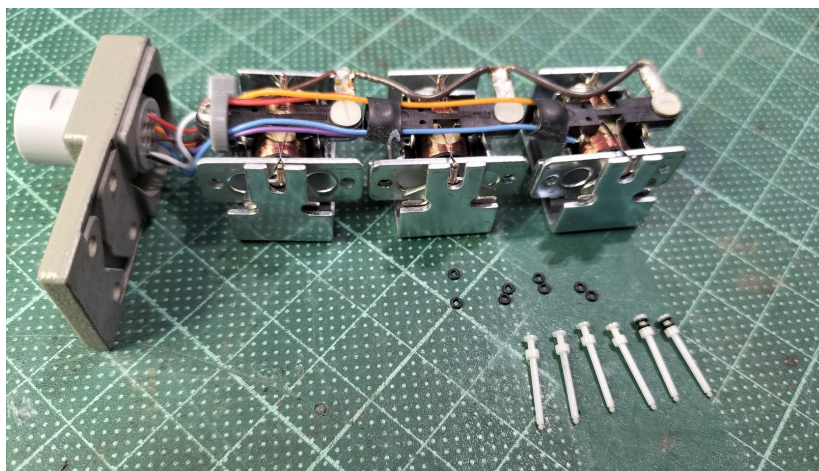
- Examine the attenuator body carefully. Mark the endplates so you know which end is which during re-assembly. Remove the lock nuts from the SMA connectors and remove the compliance plate. Under the plate you will see 10 screws that hold the attenuator assembly. Remove all the screws and pull the attenuator body away from the solenoid assembly.



- The underside of the solenoid assembly shows the plunger (blue arrow) in the correct position in a slotted hole on the switch leaf. Apply 24V to the solenoid and watch the action of the switch leaf and plunger. When you remove a solenoid, two plungers comes with it pulling the switch leaves back causing the plunger to slide out of one of the two holes (red arrows) either side of it.

Removing the solenoids

- To remove the solenoids, remove all 12 mounting screws as well as the mounting screws for the power connector. Hold the attenuator in one hand and the solenoids along with the power connector in the other. Pull the solenoids straight back from the body. The plungers will come away with the solenoids though it will take a degree force for them to pull out of the switch leaves. **Remain in control as you remove the solenoids. A sudden release of energy could damage something or send small parts flying.**
- Place the solenoids on a flat surface taking care of the wires. The plungers can now be removed from the solenoids.



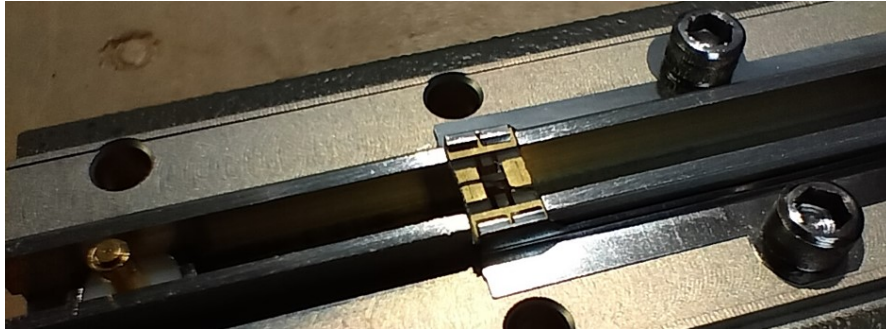
- Clean away the residue of the old o-rings and install two new rings on each plunger. Slip the plungers back in place on the solenoid actuator plates. There must be an o-ring either side of the actuator plate.
- Make sure all the plungers remain in their actuator plates and the o-rings are in the correct position as you slowly and carefully fit the solenoids back onto the attenuator block. Its tricky keeping 3 solenoids, 6 plungers and 12 o-rings in position. You will have to push a little to get everything home.
- Screw the solenoids back in position checking that the plungers and o-rings are still in place on the actuator plates. Turn the assembly over and gently move the plungers and switch leaves with tweezers until all 12 plungers are in the correct position in the slot between the two holes on their respective switch leaf.
- Cycle the power to the solenoids making sure they glide freely and the plungers remain in place. If a solenoid does not move yet the electrical connections are ok, loosen the mounting screws a little and move the solenoid about. Cycle it again and if all is well, tighten the screws.

Reassemble and test the attenuator steps are as expected then rebuild the analyzer.
Check again that the attenuation steps are as expected.

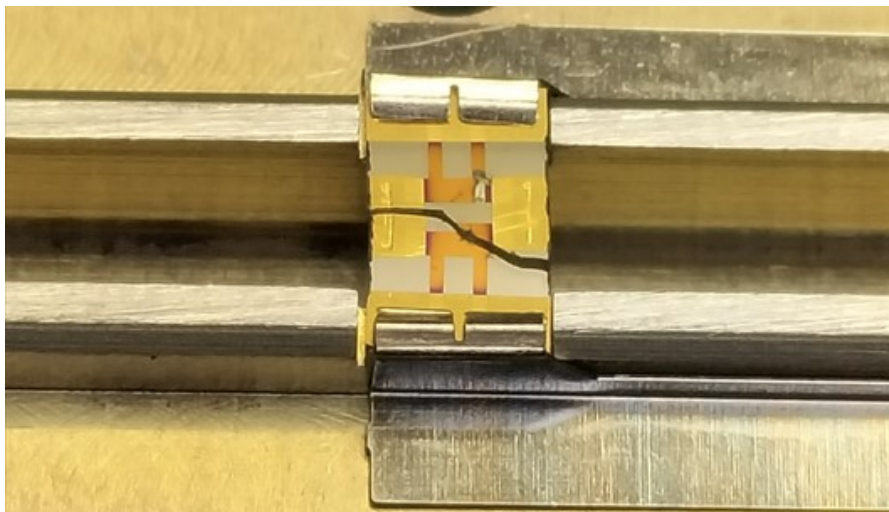
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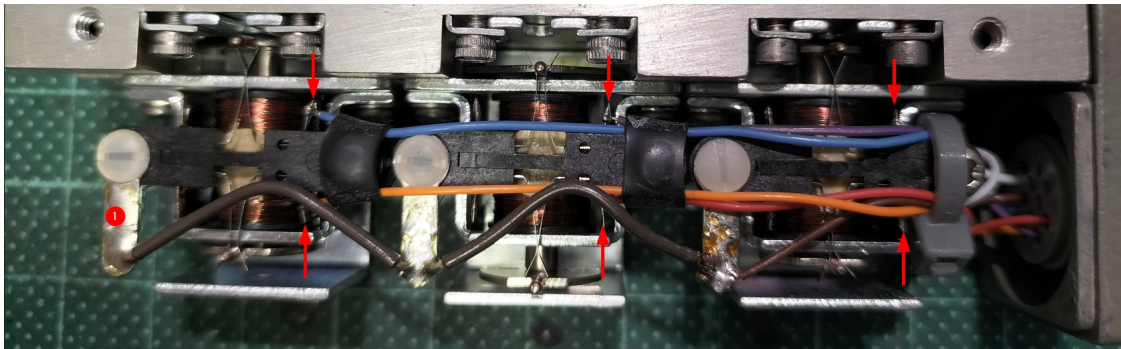
Checking the attenuator pads

- Make a visual inspection of the attenuator assembly. The attenuator pads are very delicate and easily damaged. In the image below you can see the gold SMA centre pin (left) and the 10 dB pad.



- At 5mm x 3.5mm, this pad is quite small and delicate. Finger pressure is enough to break it and the gold leaf is very soft and easily damaged by pushing on the retaining springs. In this example the pad is clearly broken.



Checking the solenoids

Be very careful: the solenoid winding wires are very thin and can easily be broken by a misplaced alligator clip or meter lead.

The solenoids require 24V at less than 10 mA. They are latching which means they only require a brief pulse to activate.

Apply +24V to the solder lug (1) and briefly ground one of the two windings as indicated by the arrows. One winding drives the plunger in, the other drives it out.